Lesson Plan

Java Pattern-1

Pre-requisites:

- Java Syntax
- Loops

List of patterns involved:

1. Solid Square
2. Solid Rectangle
3. Number Square
4. Star Triangle
5. Star Triangle Reverse
6. Number Triangle
7. Odd Number Triangle
8. Alphabet Square
9. Star Plus
10. Star Cross
11. Floyd's Triangle
12. Binary Triangle
13. Star Triangle Flipped
14. Number Triangle Flipped

Q1) Print the given Pattern:

Solid Square

```
public class Main {
    public static void main(String[] args) {
        int n = 4; // You can change this value to adjust
the size of the square

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                System.out.print("* ");
            }
            System.out.println(); // Move to the next line
after each row
        }
    }
}
```

Q2) Print the given Pattern:

Solid Rectangle

```
public class Main {
    public static void main(String[] args) {
        int rows = 4; // You can change this value to adjust
the number of rows

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < 5; j++) {
                System.out.print("* ");
            }
            System.out.println(); // Move to the next line
after each row
        }
    }
}
```

Q3) Print the given Pattern:

Number Square


```
public class Main {
    public static void main(String[] args) {
        int rows = 4;
        int columns = 4;

        for (int i = 0; i < rows; i++) {
            for (int j = 1; j <= columns; j++) {
                System.out.print(j + " ");
            }
            System.out.println(); // Move to the next line
after each row
        }
    }
}
```

Q4) Print the given Pattern:

Star Triangle

```java
public class Main {
    public static void main(String[] args) {
        int rows = 4;

        for (int i = 1; i <= rows; i++) {
            // Print '*' i times in each row
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Q5) Print the given Pattern:

Star Triangle Reverse

```java
public class Main {
    public static void main(String[] args) {
        int rows = 4;

        for (int i = rows; i >= 1; i--) {
            // Print '*' i times in each row
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Q6) Print the given Pattern:

Number Triange


```java
public class Main {
    public static void main(String[] args) {
        int rows = 4;

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

Q7) Print the given Pattern:

Odd Number Triangle


```java
public class Main {
    public static void main(String[] args) {
        int rows = 4;

        for (int i = 0; i < rows; i++) {
            int num = 1;
            for (int j = 0; j <= i; j++) {
                System.out.print(num + " ");
                num += 2;
            }
            System.out.println();
        }
    }
}
```

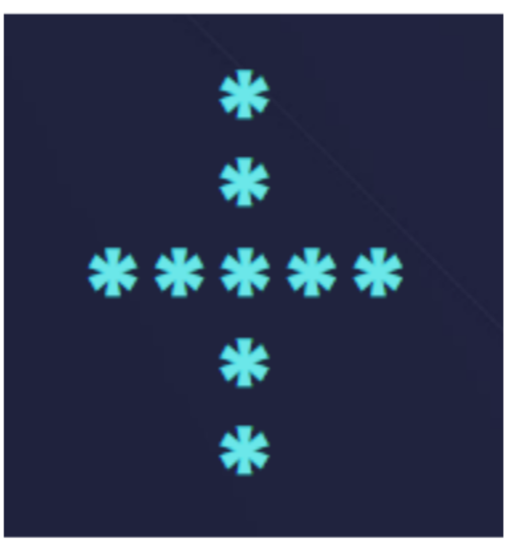
Q8) Print the given Pattern:

Alphabet Square


```java
public class Main {
    public static void main(String[] args) {
        int rows = 4;

        for (int i = 0; i < rows; i++) {
            for (char ch = 'A'; ch < 'A' + rows; ch++) {
                System.out.print(ch + " ");
            }
            System.out.println();
        }
    }
}
```

Q9) Print the given Pattern:

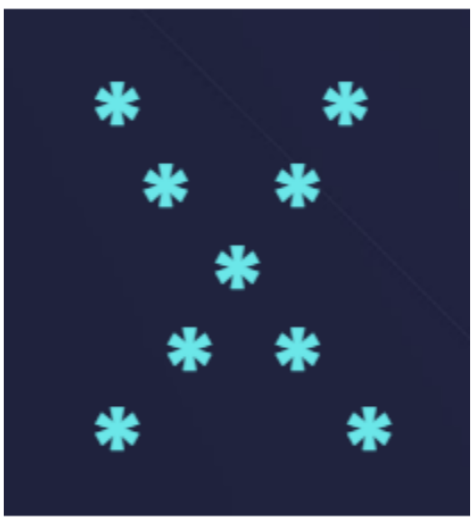
Star Plus



```
public class Main {
    public static void main(String[] args) {
        int rows = 5;

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < rows; j++) {
                if (i == 2 || j == 2) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

Q10) Print the given Pattern:

Star Cross



```
public class Main {
    public static void main(String[] args) {
        int size = 5;

        for (int i = 0; i < size; i++) {
            for (int j = 0; j < size; j++) {
                if (i == j || i + j == size - 1) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
```

```
                }
            }
            System.out.println();
        }
    }
}
```

Q11) Print the given Pattern:

Floyd's Triangle

```
1
2 3
4 5 6
7 8 9 10
```

```
public class Main {
    public static void main(String[] args) {
        int rows = 4; // You can change this value to adjust
the number of rows
        int number = 1;

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(number + " ");
                number++;
            }
            System.out.println();
        }
    }
}
```

Q12) Print the given Pattern:

Binary Triangle


```java
public class Main {
    public static void main(String[] args) {
        int rows = 4;

        for (int i = 0; i < rows; i++) {
            int value = (i+1) % 2; // Start with 1 for even rows, 0 for odd rows
            for (int j = 0; j <= i; j++) {
                System.out.print(value + " ");
                value = 1 - value; // Toggle between 0 and 1
            }
            System.out.println();
        }
    }
}
```

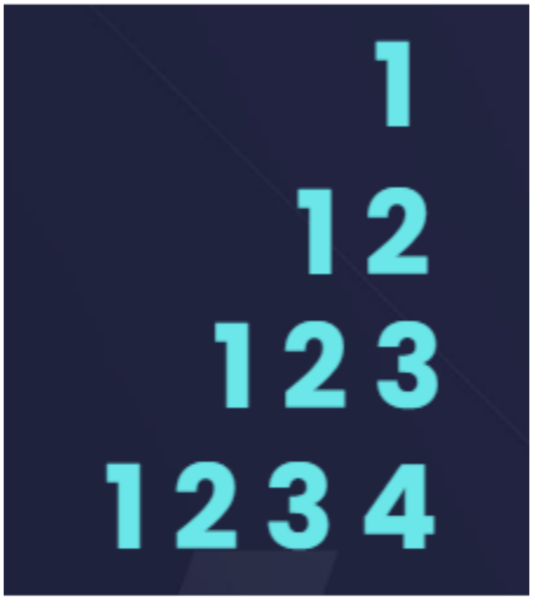
Q13) Print the given Pattern:

Star Triangle Flipped

```java
public class Main {
    public static void main(String[] args) {
        int rows = 4; // You can change this value to adjust
the number of rows

        for (int i = rows - 1; i >= 0; i--) {
            for (int j = 0; j < i; j++) {
                System.out.print("  ");
            }

            for (int k = i; k < rows; k++) {
                System.out.print("* ");
            }

            System.out.println();
        }
    }
}
```

Q14) Print the given Pattern:

Number Triangle Flipped




```java
public class Main {
    public static void main(String[] args) {
        int rows = 4; // You can change this value to adjust
the number of rows

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= rows - i; j++) {
                System.out.print("  ");
            }

            for (int k = 1; k <= i; k++) {
                System.out.print(k + " ");
            }

            System.out.println();
        }
    }
}
```

Upcoming Lecture:

More interesting patterns